Model-Agnostic Interpretation of Machine Learning via Average Marginal Effects

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***Abstract*—We propose a framework for interpreting any supervised machine learning model through the lens of average marginal effects, a standard tool in the econometrics of nonlinear models. The framework is agnostic to the choice of learner — the same estimation procedure applies whether the prediction function is a neural network, a gradient boosted model, a random forest, or any other supervised learner. Marginal effects are estimated via adaptive centered finite differences and averaged over the empirical distribution of the covariates, producing a coefficient table directly analogous to the output of a parametric regression. Standard errors are obtained via a nonparametric bootstrap that refits the model on each replicate from a fresh initialization at fixed hyperparameters, capturing model-fitting variability as well as estimation uncertainty. The framework nests the classical generalized linear model result as a special case. Empirical validation on the UCI Adult Income and Ames Housing datasets and a Monte Carlo simulation study demonstrate that average marginal effects are stable across model classes, that the bootstrap standard errors reflect how far each learner's fitted function moves under resampling, and that existing interpretability methods including SHAP lack the inferential apparatus the proposed framework provides.**

I. Introduction

The rise of flexible machine learning methods has created a persistent tension in empirical research. Models such as random forests [1], gradient boosted trees [2], and deep neural networks [3] routinely achieve predictive accuracy that parametric models cannot match. Yet the standard tools of statistical inference apply naturally only to parametric models whose functional form is specified in advance. [4] framed this as a conflict between two cultures of statistical modeling: one oriented toward interpretable data-generating models, the other toward accurate algorithmic prediction. The conventional resolution treats accuracy and interpretability as opposing ends of a spectrum, accepting reduced interpretability as the price of improved prediction. This paper argues that the tradeoff is not fundamental — it is an artifact of requiring parametric assumptions to make interpretation rigorous, and it disappears once that requirement is dropped.

The machine learning literature has produced a rich set of post-hoc interpretability methods that attempt to explain black-box model predictions without parametric assumptions. Partial dependence plots [5] visualize the marginal relationship between a feature and the predicted outcome. LIME [6] fits a local linear approximation to the prediction function around individual observations. SHAP [7] assigns each feature a contribution based on Shapley values from cooperative game theory. These tools are widely used in practice and surveyed in [8] and [9]. However, they share a fundamental limitation: they produce point estimates with no associated sampling distributions. A SHAP value, a PDP slope, and a LIME coefficient are descriptive quantities — there is no standard error, no confidence interval, and no basis for a hypothesis test. SHAP values carry a further difficulty: they measure additive attribution of a prediction relative to a baseline, not the rate of change of the prediction in a feature, so they answer a different question than the marginal effect and can diverge from it in both magnitude and sign even when a feature has a clear monotone effect on the outcome. For applied researchers who need to draw defensible conclusions from their models, these are serious constraints.

Two prior contributions are closest in spirit to the present paper. [10] propose forward marginal effects for nonlinear prediction functions as a model-agnostic interpretability method, but their framework is descriptive rather than inferential and explicitly argues against summarizing feature effects in a single metric such as the average marginal effect. The present paper takes the opposite position. [11] provide the `fmeffects` R package implementing this descriptive framework, demonstrating practical demand for marginal-effect-style interpretation of machine learning models, but without the theoretical justification — gradient consistency, bootstrap validity, characterization of when marginal effects are well-defined — that the present paper provides.

The econometric literature has solved exactly this inferential problem for parametric models through average marginal effects [12], [13]. For a generalized linear model with inverse link function μ and linear predictor $\eta = \mathbf{x}^\top \boldsymbol{\beta}$, the average marginal effect of variable j is $\mathrm{AME}_j = \beta_j \cdot \mathbb{E}[\mu'(\eta)]$ — the regression coefficient scaled by the average derivative of the inverse link, averaged over the empirical distribution of the linear predictor. The result is interpretable as the average change in the predicted outcome associated with a one-unit change in x_j, with standard errors obtained by the delta method or bootstrap. This is the output of `marginaleffects` in R and Python [14], and it is the standard form in which marginal effects are reported in applied economics, political science, sociology, and public health. The limitation is that it has been restricted to parametric models.

This paper removes that restriction. We propose a framework for computing average marginal effects for any supervised machine learning model, requiring only that the predic-

tion function $\hat{f}$ can be evaluated at a point. For models where the derivative is not available in closed form — which includes tree-based models, for which classical derivatives do not exist everywhere — we estimate it via a centered finite difference with an adaptive step size scaled to the spread of each feature. Standard errors are obtained via a nonparametric bootstrap [15], [16] that refits the model on each bootstrap sample from a fresh initialization at fixed hyperparameters, capturing model-fitting variability as well as estimation uncertainty. The result is a coefficient table — estimate, standard error, and significance level — directly analogous to the output of a parametric regression but with no restrictions on the functional form of $\hat{f}$. The framework nests the classical GLM result as a special case: when $\hat{f}$ is a logistic regression, the estimator recovers the classical AME automatically. When $\hat{f}$ is XGBoost or a random forest or a neural network, the same formula applies and produces output on the same scale and with the same interpretation. Asymptotic results establishing gradient consistency for the proposed estimator across neural networks, gradient boosted models, and Mondrian forests are developed in a companion journal paper.

We validate the framework empirically on the UCI Adult Income [17], [18] and Ames Housing [19] datasets, comparing AMEs from logistic and linear regression, random forests [1], XGBoost [2], and neural networks against SHAP values [7] and partial dependence plot slopes [5]. The AME and PDP estimates are in close agreement throughout, while SHAP values are on a different scale and in several cases have the wrong sign. Because AMEs come with standard errors and SHAP values do not, the AME framework supports conclusions that the SHAP framework cannot.

The remainder of the paper is organized as follows. Section 2 defines the estimand, presents the estimator, and describes the bootstrap inference procedure. Section 3 presents the empirical application. Section 4 presents simulation evidence on finite-sample performance. Section 5 concludes.

II. Method

A. Estimand

Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ denote the feature vector for observation i and let $y_i \in \mathbb{R}$ denote the outcome. We observe n independent draws $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ from some population distribution P. Let $f : \mathbb{R}^p \to \mathbb{R}$ denote the conditional expectation function $f(\mathbf{x}) = E[Y \mid \mathbf{x}]$. For binary outcomes this is the conditional probability of the positive class; for continuous outcomes it is the conditional mean directly. The method requires no modification across outcome types.

The marginal effect of variable j at observation i is the partial derivative of f evaluated at $\mathbf{x}_i$, and the population average marginal effect is its expectation over the marginal distribution of $\mathbf{x}$:

$$\mathrm{AME}_j = \mathbb{E}\left[\frac{\partial f}{\partial x_j}(\mathbf{x})\right] \tag{1}$$

For parametric models this estimand reduces to the classical result [12], [13]. For a generalized linear model with inverse link μ and linear predictor $\eta = \mathbf{x}^\top \boldsymbol{\beta}$, $\mathrm{AME}_j = \beta_j \cdot \mathbb{E}[\mu'(\eta)]$ — the OLS coefficient under the identity link, the average logistic density under the logistic link, and the average normal density under the probit link. The framework therefore nests the entire GLM family as a special case and reduces to the output of `marginaleffects` in R and Python [14] when the model is parametric.

B. Estimation

The estimator combines a sample average over observations with a finite-difference approximation of the partial derivative for each feature. Given an estimator $\hat{f}$ of f, the sample analog of the AME is:

$$\widehat{\mathrm{AME}}_j = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial \hat{f}}{\partial x_j}\bigg|_{\mathbf{x}_i} \tag{2}$$

For most supervised learners the partial derivative is not available in closed form, and for tree-based models it does not exist everywhere in the classical sense. We estimate it via a centered finite difference:

$$\frac{\partial \hat{f}}{\partial x_j}\bigg|_{\mathbf{x}_i} \approx \frac{\hat{f}\left(\mathbf{x}_i^{(j,\, x_{ij}+h_j)}\right) - \hat{f}\left(\mathbf{x}_i^{(j,\, x_{ij}-h_j)}\right)}{2h_j} \tag{3}$$

where $\mathbf{x}_i^{(j,v)}$ denotes $\mathbf{x}_i$ with the j-th component replaced by v. The centered difference has approximation error $O(h_j^2)$ rather than $O(h_j)$. The step size is chosen adaptively as $h_j = \max(10^{-4}, 0.05 \cdot \hat{\sigma}_j)$, where $\hat{\sigma}_j$ is the sample standard deviation of feature j. Scaling to the spread of x_j ensures the finite difference is neither numerically unstable nor too coarse. For integer-valued features the step is floored at one half, so the perturbation always reaches an adjacent level: tree ensemble predictions are piecewise constant, and a step that never crosses a split threshold returns a derivative of exactly zero.

For binary or categorical variables the partial derivative does not exist and the finite difference is replaced by the contrast $m_{ij} = \hat{f}(\mathbf{x}_i^{(j,1)}) - \hat{f}(\mathbf{x}_i^{(j,0)})$, with interpretation identical to the continuous case. For binary classification models, $\hat{f}$ must return the estimated conditional probability $\hat{P}(Y = 1 \mid \mathbf{x})$ rather than the predicted class label, available as `predict_proba(X)[:,1]` in scikit-learn and equivalent interfaces elsewhere.

C. Inference

Standard errors are obtained via a nonparametric bootstrap [15], [16]. Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ denote the observed sample and let λ denote the hyperparameters of $\hat{f}$, which are held fixed throughout. For $b = 1, \ldots, B$, draw a bootstrap sample $\mathcal{D}^{(b)}$ by sampling $(\mathbf{x}_i, y_i)$ pairs jointly with replacement, refit $\hat{f}^{(b)}$ on $\mathcal{D}^{(b)}$ from a fresh initialization at those same hyperparameters, and compute $\widehat{\mathrm{AME}}_j^{(b)}$ using $\hat{f}^{(b)}$ evaluated at $\mathcal{D}^{(b)}$. The bootstrap standard error is the standard deviation of $\{\widehat{\mathrm{AME}}_j^{(b)}\}_{b=1}^B$.

The complete procedure is summarized in Algorithm 1. Sampling pairs jointly preserves the joint distribution of features and outcome that $\hat{f}$ is trained to approximate. Each replicate is refit from a fresh initialization rather than warm-started from $\hat{f}$, so that the spread across replicates reflects how far the fitted function actually moves with the data: for the neural network, early stopping is re-run on the resample, and for the tree ensembles the full complement of trees or boosting rounds is regrown. Each replicate is evaluated on the sample that produced it, and the adaptive step sizes h_j are recomputed there, so the procedure resamples the empirical distribution the average marginal effect is taken over as well as the fit [16]. Because λ is held fixed, these standard errors reflect estimation and model-fitting variability at a given specification rather than uncertainty about the specification itself.

Algorithm 1 AME estimation with bootstrap inference

Require: Sample $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, model class, hyperparameters λ, bootstrap replicates B

Ensure: AME estimates $\{\widehat{\mathrm{AME}}_j\}_{j=1}^p$ and standard errors $\{\widehat{\mathrm{se}}_j\}_{j=1}^p$

Fit $\hat{f}$ on $\mathcal{D}$ with hyperparameters λ
for $j = 1, \ldots, p$ **do**
  if x_j continuous **then**
    Compute $\widehat{\mathrm{AME}}_j$ via centered finite difference with step size h_j
  else
    Compute $\widehat{\mathrm{AME}}_j$ via the contrast $\hat{f}(\mathbf{x}_i^{(j,1)}) - \hat{f}(\mathbf{x}_i^{(j,0)})$
  end if
end for
for $b = 1, \ldots, B$ **do**
  Draw $\mathcal{D}^{(b)}$ by sampling pairs from $\mathcal{D}$ with replacement
  Refit $\hat{f}^{(b)}$ on $\mathcal{D}^{(b)}$ from a fresh initialization at λ
  for $j = 1, \ldots, p$ **do**
    Compute $\widehat{\mathrm{AME}}_j^{(b)}$ using $\hat{f}^{(b)}$ evaluated at $\mathcal{D}^{(b)}$
  end for
end for
Set $\widehat{\mathrm{se}}_j$ to the standard deviation of $\{\widehat{\mathrm{AME}}_j^{(b)}\}_{b=1}^B$ for each j

III. Empirical Examples

Across applications we fit four model classes: logistic or linear regression as the parametric baseline, a random forest, XGBoost, and a feedforward neural network. The random forest uses 200 trees with maximum depth 8. XGBoost uses 200 boosting rounds with maximum depth 4 and learning rate 0.05. For binary classification, the neural network has two hidden layers of 64 and 32 units trained with Adam at learning rate 10^{-3}, with early stopping on a 10% validation split. For continuous regression, the network is deeper — four hidden layers of 256, 128, 64, and 32 units — trained at learning rate 3×10^{-4} with the input normalization layer and target standardization wrapper described in Section 2. Bootstrap standard errors use 200 replicates.

A. UCI Adult Income

The first dataset is the UCI Adult Income dataset [17], [18], comprising 48,842 observations of US adults. The outcome is whether an individual earns more than $50,000 per year. Features are organized into demographics (age, sex, years of education), work characteristics (hours worked per week, occupation group, work class), and financial variables (capital gains, capital losses, marital status). The specification search considers logistic regression and XGBoost across four specifications.

Table I presents the specification search. The demographic panel tells a consistent story across specifications and both model classes. Age and education have positive effects on the probability of earning above $50,000. The female indicator is large and negative, falling from -0.16 to -0.03 as financial and work controls are added — a pattern of omitted variable bias that is clearly legible in the AME framework and consistent across logistic regression and XGBoost. The XGBoost standard errors are generally the larger of the two: only a third of the coefficients are estimated at least as precisely as by logistic regression, and the median ratio runs from 1.1 to 2.8 across specifications. This is the price of the additional flexibility, since a resampled gradient boosted fit moves further from one sample to the next than a resampled linear index does. The capital gains and losses AMEs are near zero in magnitude because the displayed coefficients reflect a one-dollar change in a highly skewed variable.

Table II compares all four model classes on the full specification. The most important finding is the stability of the core demographic results across architectures. The education AME is 0.031 for logistic regression, 0.029 for the random forest, 0.031 for XGBoost and 0.028 for the neural network — a range of three tenths of a percentage point — and the married AME runs from 0.236 to 0.275. This is the kind of cross-model validation the AME framework makes possible, because it puts every learner's output on a scale where the numbers mean the same thing. The one coefficient on which the models visibly disagree is the female indicator, and the disagreement is confined to XGBoost: logistic regression, the forest and the network report -0.027, -0.022 and -0.027, while XGBoost reports -0.012, outside the logistic interval of $[-0.034, -0.019]$. The network's apparent agreement should not be read as confirmation: its standard error there is roughly three times the logistic model's, wide enough to cover the XGBoost estimate too. Its intervals are the widest in the table throughout, which is what a model whose fit depends on a stochastic optimizer should produce once the bootstrap refits it honestly.

Table III compares AMEs against SHAP values and partial dependence plot slopes. The contrast is stark. The AME and PDP estimates are in close agreement throughout — the married AME of 0.275 for logistic regression matches the PDP slope of 0.275, and similar agreement holds for the

TABLE I
UCI ADULT INCOME: SPECIFICATION SEARCH. AVERAGE MARGINAL EFFECTS ON P(INCOME > $50K).

	Logistic				XGBoost			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
Panel A: Demographics								
age	0.0060***	0.0060***	0.0024***	0.0027***	0.0059***	0.0059***	0.0029***	0.0031***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
female	-0.1730***	-0.1648***	-0.0275***	-0.0269***	-0.1592***	-0.1446***	-0.0138***	-0.0122***
	(0.0032)	(0.0033)	(0.0038)	(0.0039)	(0.0033)	(0.0037)	(0.0034)	(0.0036)
education_num	0.0523***	0.0393***	0.0400***	0.0306***	0.0610***	0.0452***	0.0405***	0.0312***
	(0.0007)	(0.0008)	(0.0006)	(0.0007)	(0.0041)	(0.0041)	(0.0041)	(0.0039)
Panel B: Work Characteristics								
hours_per_week	–	0.0048***	–	0.0033***	–	0.0084**	–	0.0032
		(0.0001)		(0.0001)		(0.0033)		(0.0020)
government	–	0.0029	–	0.0033	–	-0.0067*	–	-0.0003
		(0.0047)		(0.0042)		(0.0039)		(0.0022)
self_employed	–	-0.0077	–	-0.0268***	–	0.0113**	–	-0.0055*
		(0.0049)		(0.0042)		(0.0056)		(0.0030)
white_collar	–	0.0666***	–	0.0541***	–	0.0466***	–	0.0474***
		(0.0056)		(0.0049)		(0.0047)		(0.0039)
blue_collar	–	-0.0264***	–	-0.0270***	–	-0.0446***	–	-0.0251***
		(0.0057)		(0.0049)		(0.0054)		(0.0043)
service	–	-0.0697***	–	-0.0427***	–	-0.0807***	–	-0.0343***
		(0.0062)		(0.0057)		(0.0073)		(0.0074)
Panel C: Financial								
capital_gain	–	–	0.0000***	0.0000***	–	–	-0.0001***	-0.0001***
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
capital_loss	–	–	0.0001***	0.0001***	–	–	-0.0001***	-0.0000***
			(0.0000)	(0.0000)			(0.0000)	(0.0000)
married	–	–	0.2854***	0.2752***	–	–	0.2383***	0.2357***
			(0.0042)	(0.0040)			(0.0044)	(0.0044)
McFadden R^2	0.199	0.233	0.376	0.400	0.257	0.296	0.467	0.492
Accuracy	0.797	0.806	0.841	0.847	0.805	0.818	0.864	0.872

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates are plug-in window AMEs with adaptive step sizes; standard errors in parentheses are the standard deviation across 200 nonparametric bootstrap replicates, each refit from a fresh initialization on the resample.

occupation indicators and hours per week. This agreement is expected and reassuring: both the AME and the PDP slope estimate the derivative of the prediction function, so their numerical proximity confirms that the finite-difference AME recovers the same effect the PDP describes, while adding the standard errors the PDP lacks. The one exception is capital loss, whose PDP slope is undefined because the variable is zero for nearly all observations, so the quantile-based grid used to compute the partial dependence slope collapses to a single point; the AME, which perturbs each observation individually rather than constructing a grid, remains well-defined and is reported for all variables. The SHAP values tell a different and incompatible story. They are on a different scale, and in several cases they have the wrong sign relative to both the AME and the PDP. The SHAP value for age is negative across all four models while the AME and PDP are both positive, even though older individuals are unambiguously more likely to earn above $50,000 in these data. This is not a numerical artifact but a consequence of what SHAP measures: a mean signed SHAP value summarizes how much a feature pushes predictions away from a baseline, averaged over a particular reference distribution, which is not the average rate of change of the prediction in that feature. Practitioners who read a SHAP summary as a marginal effect — the larger the value, the larger the effect on the outcome — are therefore drawing an inference the quantity does not support. And because no standard errors are reported for SHAP or PDP, a researcher using either alone has no basis to test whether age, education, or any other variable has a statistically distinguishable effect.

B. Ames Housing

The Ames Housing dataset [19] comprises approximately 1,460 observations of residential home sales in Ames, Iowa between 2006 and 2010. The outcome is sale price in dollars, so AMEs are interpreted in dollar units. Features are organized into physical characteristics, quality and condition, and lot and location indicators. The specification search considers linear regression and a random forest.

Table IV presents the specification search. Above-ground living area carries an AME of $108 per square foot in the

TABLE II
UCI ADULT INCOME: MODEL COMPARISON. AVERAGE MARGINAL EFFECTS ON P(INCOME > $50K), FULL SPECIFICATION (A+B+C).

Feature	Logistic	Random Forest	XGBoost	Neural Net
age	0.0027***	0.0042***	0.0031***	0.0039***
	(0.0001)	(0.0002)	(0.0001)	(0.0004)
female	-0.0269***	-0.0218***	-0.0122***	-0.0265**
	(0.0039)	(0.0026)	(0.0036)	(0.0123)
education_num	0.0306***	0.0294***	0.0312***	0.0281***
	(0.0007)	(0.0025)	(0.0039)	(0.0030)
hours_per_week	0.0033***	0.0070***	0.0032	0.0055***
	(0.0001)	(0.0024)	(0.0020)	(0.0007)
government	0.0033	0.0051***	-0.0003	0.0131
	(0.0042)	(0.0016)	(0.0022)	(0.0166)
self_employed	-0.0268***	0.0032**	-0.0055*	-0.0069
	(0.0042)	(0.0016)	(0.0030)	(0.0169)
white_collar	0.0541***	0.0600***	0.0474***	0.0326*
	(0.0049)	(0.0036)	(0.0039)	(0.0183)
blue_collar	-0.0270***	-0.0240***	-0.0251***	-0.0782***
	(0.0049)	(0.0027)	(0.0043)	(0.0199)
service	-0.0427***	-0.0176***	-0.0343***	-0.0479*
	(0.0057)	(0.0026)	(0.0074)	(0.0275)
capital_gain	0.0000***	0.0000***	-0.0001***	0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
capital_loss	0.0001***	-0.0000***	-0.0000***	-0.0000
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
married	0.2752***	0.2446***	0.2357***	0.2388***
	(0.0040)	(0.0044)	(0.0044)	(0.0138)
McFadden R^2	0.400	0.450	0.492	0.455
Accuracy	0.847	0.863	0.872	0.860

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates and standard errors as in Table I.

TABLE III
UCI ADULT INCOME: METHOD COMPARISON. AME, SHAP, AND PDP ESTIMATES FOR P(INCOME > $50K), FULL SPECIFICATION (A+B+C).

	Logistic			Random Forest			XGBoost			Neural Net		
	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP	AME	SHAP	PDP
age	0.0027	-0.0270	0.0028	0.0042	-0.0068	0.0024	0.0031	-0.0097	0.0036	0.0039	0.0044	0.0033
female	-0.0269	0.0022	-0.0270	-0.0218	0.0011	-0.0218	-0.0122	0.0015	-0.0122	-0.0265	-0.0001	-0.0222
education_num	0.0306	-0.1186	0.0269	0.0294	-0.0077	0.0137	0.0312	-0.0093	0.0170	0.0281	-0.0051	0.0207
hours_per_week	0.0033	-0.0691	0.0033	0.0070	-0.0054	0.0020	0.0032	-0.0080	0.0028	0.0055	-0.0032	0.0028
government	0.0033	-0.0009	0.0026	0.0051	-0.0003	0.0051	-0.0003	-0.0008	-0.0003	0.0131	0.0002	0.0082
self_employed	-0.0268	0.0016	-0.0265	0.0032	-0.0003	0.0032	-0.0055	-0.0002	-0.0055	-0.0069	-0.0008	-0.0045
white_collar	0.0541	0.0127	0.0538	0.0600	0.0031	0.0600	0.0474	0.0025	0.0474	0.0326	0.0007	0.0312
blue_collar	-0.0270	0.0136	-0.0273	-0.0240	0.0010	-0.0240	-0.0251	0.0004	-0.0251	-0.0782	-0.0008	-0.0531
service	-0.0427	-0.0027	-0.0427	-0.0176	0.0003	-0.0176	-0.0343	0.0005	-0.0343	-0.0479	0.0011	-0.0558
capital_gain	0.0000	-0.2627	0.0000	0.0000	-0.0224	0.0000	-0.0001	-0.0264	-0.0001	0.0000	0.0042	0.0000
capital_loss	0.0001	0.0027	–	-0.0000	-0.0034	–	-0.0000	-0.0062	–	-0.0000	-0.0115	–
married	0.2752	-0.0768	0.2749	0.2446	-0.0064	0.2446	0.2357	-0.0049	0.2357	0.2388	0.0149	0.1958

Notes: AME = Average Marginal Effect (marginfx). SHAP = mean signed SHAP value (GradientExplainer). PDP = slope of partial dependence plot. Full specification (A+B+C) only. No standard errors reported for SHAP and PDP.

TABLE IV
AMES HOUSING: SPECIFICATION SEARCH. AVERAGE MARGINAL EFFECTS ON SALE PRICE (USD).

	Linear				Random Forest			
	Spec A	A+B	A+C	A+B+C	Spec A	A+B	A+C	A+B+C
Panel A: Physical Characteristics								
GrLivArea	108.22***	85.99***	74.32***	65.57***	126.41***	79.37***	78.92***	56.44***
	(12.93)	(14.30)	(10.73)	(13.06)	(14.44)	(6.96)	(6.51)	(5.25)
BedroomAbvGr	-27911.62***	-10181.55***	-11440.18***	-6189.02**	-23128.28***	-3055.72***	-5528.45***	-1470.18**
	(3139.45)	(2873.76)	(2959.25)	(2879.08)	(3865.06)	(862.50)	(2029.56)	(662.67)
FullBath	30380.78***	-9793.90*	5378.67	-6287.42	18535.37***	382.25	1687.49	498.18
	(5400.68)	(5403.76)	(3551.42)	(4381.13)	(2643.64)	(974.59)	(1538.35)	(1007.14)
HalfBath	3586.62	-13111.01***	-728.29	-6967.90**	493.43	-1160.44**	420.55	-198.68
	(3637.30)	(3392.06)	(2504.64)	(2991.66)	(1535.06)	(544.86)	(428.99)	(241.35)
Panel B: Quality and Condition								
OverallQual	–	21388.25***	–	16465.22***	–	17049.34***	–	15636.08***
		(1861.48)		(1286.53)		(1558.54)		(1642.54)
OverallCond	–	5810.75***	–	6311.48***	–	2774.28***	–	1974.55***
		(1047.66)		(923.64)		(752.72)		(589.68)
YearBuilt	–	685.31***	–	440.47***	–	729.64***	–	576.01***
		(80.56)		(102.13)		(207.66)		(150.36)
YearRemodAdd	–	72.76	–	101.51*	–	756.23***	–	518.52***
		(67.05)		(56.55)		(224.23)		(167.84)
Panel C: Lot and Location								
LotArea	–	–	0.33	0.61***	–	–	1.97***	1.81***
			(0.21)	(0.23)			(0.37)	(0.33)
GarageArea	–	–	72.25***	38.68***	–	–	82.79***	50.03***
			(7.99)	(6.87)			(13.81)	(10.37)
nbhd_high	–	–	90316.66***	51433.02***	–	–	51108.94***	5058.64**
			(7629.44)	(9947.22)			(13205.57)	(2138.24)
nbhd_mid	–	–	31391.51***	7508.07	–	–	25973.59***	5949.97***
			(2981.69)	(4816.36)			(3985.75)	(1801.45)
R^2	0.584	0.758	0.739	0.802	0.829	0.937	0.926	0.953
RMSE	51,243.9	39,080.1	40,596.4	35,355.5	32,830.9	19,881.6	21,534.2	17,244.4

Notes: ***p<0.01, **p<0.05, *p<0.10. Estimates and standard errors as in Table I.

demographics-only specification for linear regression, falling to \$66 in the full specification as quality, condition, and location controls absorb variation — a pattern of positive omitted variable bias since larger homes tend to be in better neighborhoods and of higher quality. The random forest's standard errors are not uniformly tighter than the linear model's, and the pattern is informative. For above-ground living area the forest reports \$5.25 against the linear model's \$13.06, and for the high neighborhood indicator \$2,138 against \$9,947; but for the overall quality rating the ordering reverses, \$1,643 for the forest against \$1,287 for the linear model. The forest fits substantially better (R^2 of 0.953 against 0.802), and where the linear model must represent a nonlinear relationship with a single slope it pays for that in precision. Where the relationship is close to linear, as it is for the quality rating, the parametric model is the more efficient of the two and the ensemble's refitting variability shows through instead. The overall quality rating is the largest single predictor, with an AME of approximately \$21,000 per rating point for linear regression and \$17,000 for the random forest, both precisely estimated. The neighborhood tier indicators show substantial attenuation as quality controls are added — the high neighborhood AME falls from \$90,000 to \$51,000 for linear regression and from \$51,000 to \$5,000 for the random forest — suggesting that neighborhood effects are largely mediated by quality and condition variables in a way the linear model cannot fully represent.

IV. SIMULATION

We evaluate the finite-sample performance of the AME estimator through two simulation studies. Simulation 1 examines bias and root mean squared error as a function of sample size. Simulation 2 examines bootstrap standard error calibration via empirical coverage of nominal 95% confidence intervals. All features x_1, x_2, x_3, x_4 are drawn independently from $N(0, 1)$. Features x_1 and x_2 have nonzero true effects; x_3 and x_4 are pure noise with true AME of zero. The linear DGP sets $\eta = 2x_1 + 3x_2$. For regression, $y = \eta + \varepsilon$ with $\varepsilon \sim N(0, 1)$; for classification, $P(y = 1) = \sigma(\eta)$ where σ denotes the logistic function. True AMEs are computed via Monte Carlo integration using $n = 1{,}000{,}000$ observations drawn from the true prediction function directly. Sample sizes of $n \in \{250, 500, 1\,000, 2\,500, 5\,000\}$ are evaluated over 500 Monte Carlo iterations for classification and 1,000 iterations for regression. Nonlinear and interaction DGPs are evaluated in the journal version.

The simulation fits four model classes. The random forest uses 100 trees with maximum depth 6. XGBoost uses 100 boosting rounds with maximum depth 4 and learning rate

0.05. The neural network is a feedforward architecture with two hidden layers of 32 and 16 units, trained for 50 epochs with batch size 64 and learning rate 10^{-3} using Adam. For regression outcomes, the network includes the input normalization layer and target standardization wrapper described in Section 2. Logistic and linear regression baselines use no hyperparameter tuning. Bootstrap calibration in Simulation 2 uses 1,000 iterations and 1,000 replicates for all models except the neural network, which uses 500 and 200.

Table V reports bias and RMSE for the linear classification DGP, where the true AMEs of x_1 and x_2 are 0.199 and 0.298 — not 2.0 and 3.0 because the sigmoid compresses the scale. Logistic regression recovers the true AMEs with negligible bias at all sample sizes, and RMSE declines from 0.020 at $n = 250$ to 0.005 at $n = 5{,}000$ at the expected $O(n^{-1/2})$ rate. The neural network behaves similarly. XGBoost shows larger bias at small samples — around 0.06 at $n = 250$ — reflecting underfitting, but converges by $n = 2{,}500$. Random forests show a different pattern: bias is small at $n = 250$ and $n = 500$ but turns negative and persistent at $n = 2{,}500$ and $n = 5{,}000$, reaching -0.016 at the largest sample size. This reflects regularization bias — the tree depth and subsampling constraints impose shrinkage that does not vanish with sample size at default tuning. The noise features have bias and RMSE near zero throughout for all model classes.

Table VI reports the regression setting with the linear DGP, $y = 2x_1 + 3x_2 + \varepsilon$. Linear regression recovers the true AMEs of 2.0 and 3.0 with negligible bias and RMSE declining from 0.066 at $n = 250$ to 0.014 at $n = 5{,}000$. The regression setting reveals random forest regularization bias more starkly: bias reaches -0.091 for x_1 and -0.095 for x_2 at $n = 5{,}000$, and RMSE is not declining at the largest sample sizes. XGBoost shows large positive bias at small samples — 0.41 for x_1 at $n = 250$ — but converges quickly. The neural network shows moderate negative bias at small samples that falls toward zero by $n = 2{,}500$.

Table VII reports bootstrap standard error calibration for the linear classification DGP. Logistic regression is well calibrated throughout, covering between 0.902 and 0.946 across variables and sample sizes, and its bootstrap standard error tracks the sampling deviation of the estimator to within five percent at every sample size. The neural network undercovers at $n = 250$, most visibly for x_2 at 0.812, and recovers to near nominal by $n = 1{,}000$; the shortfall is a small-sample effect, and the network's intervals drift slightly conservative thereafter. XGBoost and the random forest are conservative at the smaller sample sizes — coverage between 0.93 and 0.99 — because their bootstrap standard errors run 1.2 to 1.7 times the true sampling deviation of the estimator, so their intervals are wider than they need to be.

The random forest is the exception at $n = 5{,}000$, where coverage falls to 0.802 for x_1 and 0.890 for x_2 while remaining near nominal for the noise features. The mechanism is the regularization bias documented in Simulation 1. The forest's bias on x_1 is 0.013 at $n = 250$, -0.002 at $n = 1{,}000$ and -0.016 at $n = 5{,}000$: it does not shrink with the sample. Its standard error does, from 0.061 to 0.023 to 0.007. Measured in standard errors the bias therefore runs 0.21, 0.08 and 2.11, and coverage follows it down. At $n = 5{,}000$ the half-width of the interval, 0.0143, is smaller than the bias it would have to span, 0.0157.

What makes this diagnosis unambiguous is that the forest's standard error is not too small. It is 1.2 to 1.5 times the estimator's true sampling deviation at these sample sizes, so the intervals are if anything too wide, and the coverage failure cannot be attributed to an understated standard error. It is attributable to bias alone. XGBoost makes the same point from the other direction: its bias is larger than the forest's at $n = 250$ (0.059 against 0.013) but it vanishes with the sample, reaching -0.005 at $n = 5{,}000$, so the ratio of bias to standard error never exceeds 0.75 and coverage never degrades in the direction the forest's does. The failure is thus specific to learners whose regularization bias does not shrink, rather than to flexible learners in general. This is the empirical counterpart to the bootstrap validity conjecture for random forests: the bootstrap is consistent for the variability of $\widehat{\mathrm{AME}}_j$ around its expectation under the forest, but that expectation does not coincide with the true AME, and no amount of resampling reveals the difference.

V. Conclusion

This paper has proposed a framework for interpreting any supervised machine learning model through the lens of average marginal effects. The central claim is that the traditional accuracy-interpretability tradeoff is not fundamental. By defining the average marginal effect as the expectation of the partial derivative of the prediction function with respect to each feature, averaged over the empirical distribution of the covariates, we obtain an estimand that is well-defined for any supervised learner and reduces to the classical econometric AME when the model is a generalized linear model. The estimator replaces analytical derivatives with adaptive centered finite differences and produces a coefficient table directly analogous to the output of a parametric regression. The nonparametric bootstrap, which refits the model from scratch on each replicate, provides standard errors for these effects — an inferential apparatus that existing interpretability methods including SHAP [7], LIME [6], and partial dependence plots [5] do not provide.

The empirical applications to UCI Adult Income and Ames Housing, together with the simulation study, validate the framework. AME estimates are stable across model classes where the underlying models are well-fitted, the bootstrap standard errors track how far each learner's fitted function moves under resampling rather than its goodness of fit, and AME and PDP estimates are in close agreement throughout. The comparison with SHAP values reveals a systematic divergence: SHAP values are on a different scale, in several cases have the wrong sign, and carry no inferential apparatus. The dollar-denominated Ames Housing application makes the divergence concrete — SHAP attributes to above-ground living area an effect fifty times larger than the AME for the linear

TABLE V
SIMULATION 1: AME RECOVERY — LINEAR DGP (CLASSIFICATION). MEAN BIAS AND RMSE (IN PARENTHESES) ACROSS 500 MONTE CARLO ITERATIONS.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Logistic						
x1	0.199	-0.0058	-0.0034	-0.0014	-0.0008	-0.0006
		(0.0201)	(0.0146)	(0.0106)	(0.0064)	(0.0048)
x2	0.298	-0.0109	-0.0057	-0.0026	-0.0010	-0.0005
		(0.0229)	(0.0153)	(0.0105)	(0.0066)	(0.0046)
x3	0.000	0.0012	0.0006	0.0006	0.0003	0.0002
		(0.0198)	(0.0136)	(0.0099)	(0.0063)	(0.0045)
x4	0.000	0.0008	-0.0001	-0.0008	-0.0001	0.0003
		(0.0197)	(0.0136)	(0.0100)	(0.0062)	(0.0045)
Panel: Random Forest						
x1	0.199	0.0126	0.0044	-0.0018	-0.0111	-0.0157
		(0.0412)	(0.0239)	(0.0160)	(0.0142)	(0.0168)
x2	0.298	0.0199	0.0113	-0.0008	-0.0102	-0.0158
		(0.0500)	(0.0309)	(0.0174)	(0.0143)	(0.0172)
x3	0.000	0.0007	0.0010	0.0005	0.0002	0.0002
		(0.0284)	(0.0171)	(0.0104)	(0.0056)	(0.0034)
x4	0.000	-0.0008	-0.0013	-0.0012	-0.0001	0.0002
		(0.0267)	(0.0177)	(0.0106)	(0.0055)	(0.0034)
Panel: XGBoost						
x1	0.199	0.0590	0.0265	0.0107	-0.0016	-0.0049
		(0.0782)	(0.0401)	(0.0223)	(0.0108)	(0.0083)
x2	0.298	0.0643	0.0315	0.0085	-0.0016	-0.0046
		(0.0893)	(0.0490)	(0.0218)	(0.0116)	(0.0083)
x3	0.000	0.0008	-0.0002	0.0006	0.0002	0.0002
		(0.0332)	(0.0193)	(0.0110)	(0.0056)	(0.0033)
x4	0.000	0.0008	-0.0009	-0.0006	0.0000	0.0002
		(0.0344)	(0.0198)	(0.0116)	(0.0054)	(0.0032)
Panel: Neural Net						
x1	0.199	-0.0078	-0.0007	0.0005	0.0003	-0.0002
		(0.0222)	(0.0156)	(0.0115)	(0.0074)	(0.0058)
x2	0.298	-0.0162	-0.0017	0.0004	0.0004	0.0002
		(0.0276)	(0.0174)	(0.0116)	(0.0073)	(0.0052)
x3	0.000	0.0014	0.0010	0.0007	0.0003	0.0004
		(0.0204)	(0.0144)	(0.0103)	(0.0069)	(0.0057)
x4	0.000	0.0009	-0.0003	-0.0007	-0.0001	0.0004
		(0.0203)	(0.0143)	(0.0105)	(0.0070)	(0.0055)

Notes: Mean bias and RMSE (in parentheses) computed over 500 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

model, a number that is not interpretable as a marginal effect in any standard sense.

Several important problems remain open. A companion journal paper establishes gradient consistency for neural networks, gradient boosted models, and Mondrian forests, but bootstrap validity for the AME estimator is conjectured rather than proved across all three classes. The closest existing results — the infinitesimal jackknife for random forests [20] and the bootstrap theory for honest forests [21] — do not apply directly to the gradient functional studied here. The simulation results in Section 4 document a clear coverage failure for Breiman random forests at large sample sizes, driven by regularization bias that the standard nonparametric bootstrap does not correct, suggesting that the path to bootstrap validity may require alternatives such as Mondrian forests where the partition structure is data-independent. The companion paper takes up a further practical question with a credit default application: what the estimand means when a substantial share of the sample is trimmed, as it is for discrete features whose mass concentrates at the edges of their range, and for integer-valued features where the floored step size makes the reported effect a one-level contrast. Extension of the framework to additional model classes including support vector machines and Bayesian additive regression trees, to causal estimands beyond the predictive AME, and to computationally efficient inference procedures based on the infinitesimal jackknife or subsampling are also natural directions for future work.

TABLE VI
SIMULATION 1: AME RECOVERY — LINEAR DGP (REGRESSION). MEAN BIAS AND RMSE (IN PARENTHESES) ACROSS 1,000 MONTE CARLO ITERATIONS.

		Bias (RMSE)				
Variable	True AME	$n = 250$	$n = 500$	$n = 1,000$	$n = 2,500$	$n = 5,000$
Panel: Linear						
x1	2.000	-0.0017	-0.0001	-0.0000	-0.0006	-0.0003
		(0.0659)	(0.0445)	(0.0322)	(0.0207)	(0.0137)
x2	3.000	-0.0043	-0.0017	0.0007	-0.0004	-0.0003
		(0.0636)	(0.0453)	(0.0316)	(0.0197)	(0.0140)
x3	0.000	-0.0001	-0.0006	-0.0005	0.0012	-0.0003
		(0.0650)	(0.0457)	(0.0312)	(0.0199)	(0.0140)
x4	0.000	-0.0034	0.0006	-0.0002	-0.0003	-0.0001
		(0.0642)	(0.0452)	(0.0315)	(0.0196)	(0.0139)
Panel: Random Forest						
x1	2.000	-0.0973	-0.0332	-0.0387	-0.0703	-0.0911
		(0.2064)	(0.1189)	(0.0850)	(0.0824)	(0.0956)
x2	3.000	-0.0215	-0.0105	-0.0433	-0.0797	-0.0953
		(0.2338)	(0.1456)	(0.0997)	(0.0958)	(0.1025)
x3	0.000	-0.0015	-0.0005	-0.0005	0.0001	0.0000
		(0.0562)	(0.0290)	(0.0137)	(0.0036)	(0.0011)
x4	0.000	-0.0021	-0.0010	-0.0005	-0.0002	0.0000
		(0.0572)	(0.0297)	(0.0138)	(0.0037)	(0.0011)
Panel: XGBoost						
x1	2.000	0.4114	0.1780	0.0366	-0.0337	-0.0467
		(0.4644)	(0.2227)	(0.0917)	(0.0554)	(0.0547)
x2	3.000	0.3122	0.0948	-0.0172	-0.0558	-0.0601
		(0.4103)	(0.1884)	(0.0947)	(0.0754)	(0.0677)
x3	0.000	0.0007	0.0016	-0.0003	0.0011	-0.0001
		(0.1090)	(0.0550)	(0.0313)	(0.0153)	(0.0083)
x4	0.000	-0.0080	0.0002	0.0011	-0.0006	0.0001
		(0.1077)	(0.0597)	(0.0314)	(0.0149)	(0.0081)
Panel: Neural Net						
x1	2.000	-0.0659	-0.0208	-0.0063	-0.0021	-0.0008
		(0.1161)	(0.0544)	(0.0344)	(0.0293)	(0.0329)
x2	3.000	-0.1231	-0.0349	-0.0102	-0.0031	-0.0006
		(0.1765)	(0.0638)	(0.0361)	(0.0324)	(0.0372)
x3	0.000	0.0014	-0.0006	-0.0003	0.0009	-0.0006
		(0.0788)	(0.0484)	(0.0330)	(0.0250)	(0.0280)
x4	0.000	-0.0002	0.0008	-0.0003	-0.0004	-0.0017
		(0.0752)	(0.0496)	(0.0336)	(0.0247)	(0.0287)

Notes: Mean bias and RMSE (in parentheses) computed over 1,000 Monte Carlo iterations. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

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TABLE VII
SIMULATION 2: BOOTSTRAP SE CALIBRATION — LINEAR DGP (CLASSIFICATION). NOMINAL COVERAGE TARGET: 95%. COVERAGE RATE AND MEAN 95% CI WIDTH (IN PARENTHESES) ACROSS MONTE CARLO ITERATIONS; SEE NOTES FOR COUNTS BY MODEL.

		Coverage (CI Width)		
Variable	True AME	$n = 250$	$n = 1,000$	$n = 5,000$
Panel: Logistic				
x1	0.199	0.934	0.930	0.924
		(0.0743)	(0.0390)	(0.0174)
x2	0.298	0.902	0.940	0.946
		(0.0763)	(0.0400)	(0.0181)
x3	0.000	0.932	0.928	0.946
		(0.0737)	(0.0379)	(0.0170)
x4	0.000	0.932	0.940	0.936
		(0.0739)	(0.0381)	(0.0170)
Panel: Random Forest				
x1	0.199	0.992	0.978	0.802
		(0.2354)	(0.0872)	(0.0286)
x2	0.298	0.980	0.982	0.890
		(0.2740)	(0.1040)	(0.0343)
x3	0.000	0.996	0.970	0.962
		(0.1545)	(0.0544)	(0.0158)
x4	0.000	0.988	0.978	0.952
		(0.1537)	(0.0545)	(0.0159)
Panel: XGBoost				
x1	0.199	0.938	0.888	0.994
		(0.3055)	(0.1102)	(0.0324)
x2	0.298	0.926	0.946	0.998
		(0.3591)	(0.1289)	(0.0377)
x3	0.000	0.992	0.980	0.960
		(0.1699)	(0.0606)	(0.0162)
x4	0.000	0.980	0.978	0.956
		(0.1721)	(0.0610)	(0.0163)
Panel: Neural Net				
x1	0.199	0.894	0.938	0.982
		(0.0803)	(0.0444)	(0.0231)
x2	0.298	0.812	0.956	0.964
		(0.0879)	(0.0487)	(0.0225)
x3	0.000	0.942	0.946	0.980
		(0.0759)	(0.0397)	(0.0227)
x4	0.000	0.940	0.938	0.984
		(0.0765)	(0.0398)	(0.0227)

Notes: Coverage rate and mean 95% CI width (in parentheses). Intervals are bootstrap percentile intervals. Logistic, random forest and XGBoost use 1,000 Monte Carlo iterations and 1,000 resamples; the neural network uses 500 and 200, since every one of its resamples is a fresh training run. True AMEs computed via Monte Carlo integration with $n = 1{,}000{,}000$ observations. Noise features x3 and x4 have true AME of zero.

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